

Introduction to MLflow

A comprehensive guide to managing your machine learning lifecycle



What Is MLflow?

MLflow is an **open-source platform** designed to manage the complete machine learning lifecycle from experimentation to production deployment.

Works with any ML library

Compatible with PyTorch, TensorFlow, Scikit-learn, and other popular frameworks

Flexible deployment

Supports both local development environments and cloud-based infrastructures

Industry standard

Widely adopted across industry and research organizations



Core Components of MLflow

MLflow's modular architecture lets you use all components together or individually based on your needs

1

Tracking

Log and query experiments: metrics, parameters, code versions, and artifacts

2

Projects

Package ML code in a reusable, reproducible form for sharing with other data scientists

3

Models

Package ML models in a standard format for a variety of serving environments

4

Model Registry

Store, annotate, discover, and manage models in a central repository

MLflow Tracking

The **cornerstone of reproducible machine learning** that automatically logs everything you need:

- Hyperparameters used for training
- Performance metrics (accuracy, loss, F1-score)
- Artifacts (model files, visualizations, feature importance)
- Environment details (Python version, library dependencies)

All accessible through an intuitive UI and programmatic API



Example – Logging with MLflow

Track your experiments with just a few lines of code

```
import mlflow

mlflow.start_run()
mlflow.log_param("lr", 0.01)
mlflow.log_metric("accuracy", 0.92)
mlflow.log_artifact("model.pth")
mlflow.end_run()
```

✓ That's it!

With just 3 lines of code, you've created a fully tracked experiment

92%

Accuracy

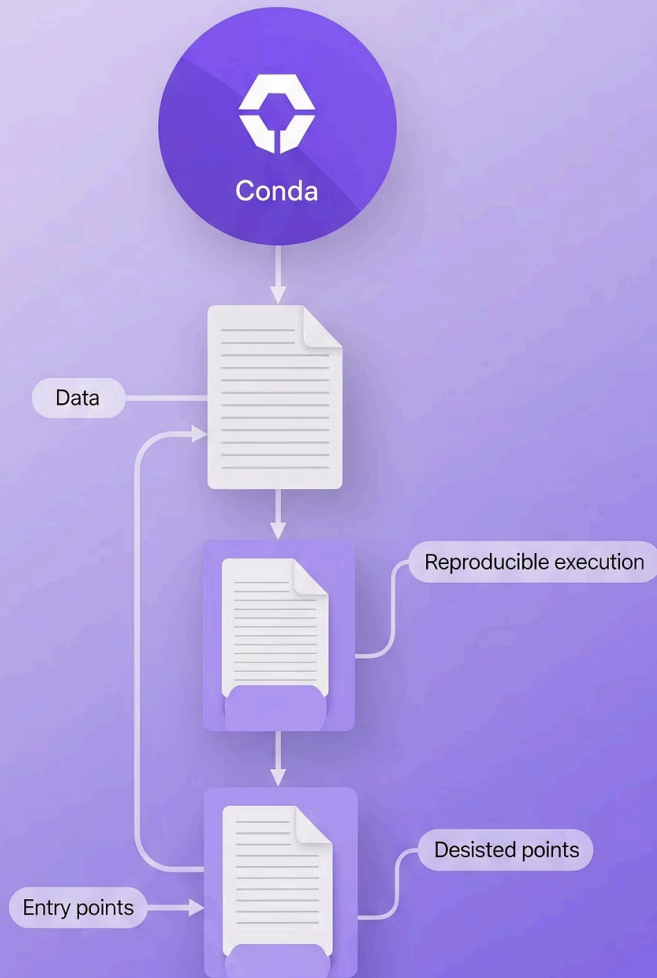
Tracked automatically

0.01

Learning Rate

Parameter logged

MLflow project



MLflow Projects

Define reproducible ML workflows that can be shared and executed

Environment Management

Uses `conda.yaml` or `requirements.txt` to ensure consistent dependencies

Standardized Entry Points

Define clear interfaces for training and inference processes

Git Integration

Seamlessly works with version control for tracking code changes

MLflow Models

A standard format for packaging machine learning models that works with diverse serving tools

Supported "Flavors"

TensorFlow

PyTorch

Scikit-learn

ONNX



Models can be deployed to multiple platforms with minimal configuration

MLflow Model Registry

1

Version Control

Track model iterations (v1, v2, v3) with complete lineage information

2

Stage Transitions

Move models through **Staging** → **Production** → **Archived** lifecycle

3

Metadata Tracking

Record who trained models, which datasets were used, and performance metrics

The foundation for enterprise MLOps and governance

MLflow Model Registry

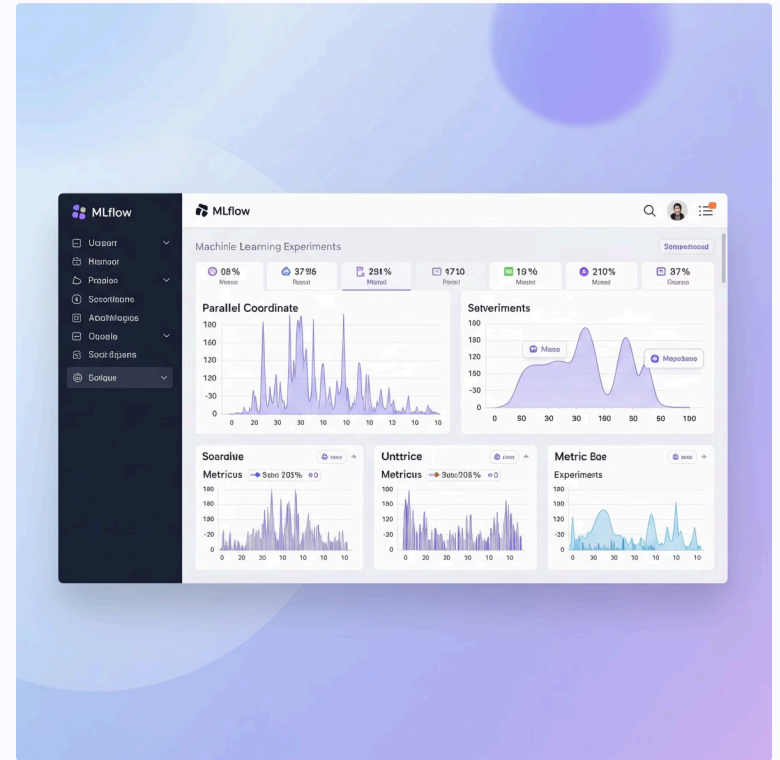
Model Name	Version	Stage	Last Modified	Use
		None		
v1 FraudDetectionModel	v1	Staging	None	Archived
v2 CustomerChurnPredictor	v2	Staging	None	Production
v3 ImageClassifierV2	v2	Staging	None	Archived
v7	v3	Production	None	Archived
v6 RecommendationEngine	v3	Archived	1BAS	Archived
v4	v3	Archived	OPLS	

MLflow UI

Intuitive web interface that enables:

- Browsing all experiment runs
- Comparing metrics and visualizations across multiple experiments
- Searching and filtering by parameters
- Examining artifacts and model details

Useful for both individual researchers and teams collaborating on projects



Can be deployed locally on your laptop or hosted on cloud infrastructure

Benefits of MLflow



Framework Agnostic

Works with all major ML frameworks and libraries, avoiding vendor lock-in



Lightweight Integration

Minimal code changes required to add tracking to existing ML workflows



Team Collaboration

Shared experiment results and reproducible workflows improve coordination



Regulatory Compliance

Experiment tracking provides audit trails for model development decisions



Research to Production

Bridges the gap between experimentation and deployment environments

Key Takeaways

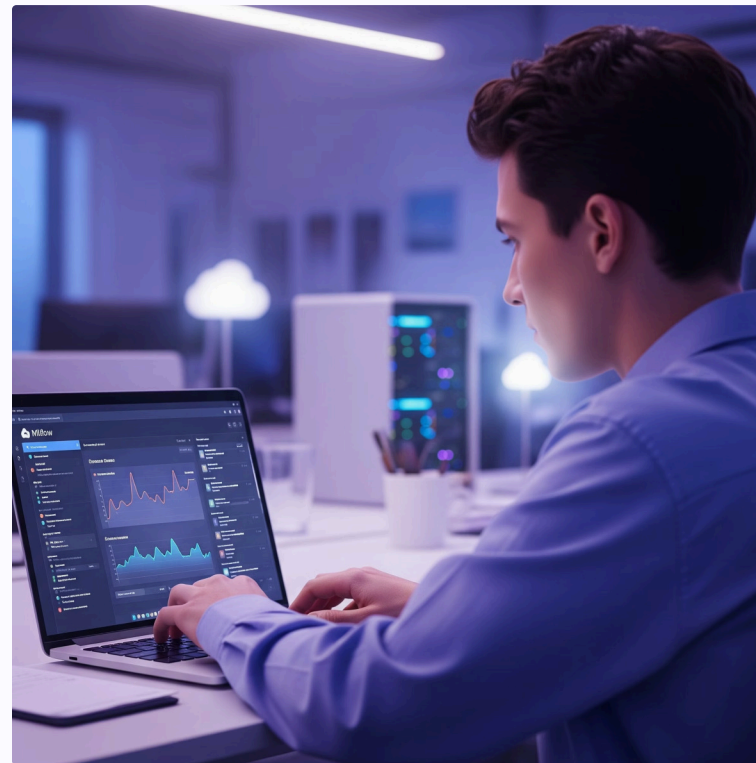
MLflow is a comprehensive open-source platform for managing the entire machine learning lifecycle

Built on **four core components**: Tracking, Projects, Models, and Registry

Dramatically **simplifies experiment tracking** and model deployment with minimal code changes

Scales seamlessly from **local development to enterprise cloud** environments

An **essential tool** in the modern AI infrastructure engineer's toolkit



Get started today at mlflow.org